Fall 2025 ECEN 629 Homework Set 4

Due Nov. 8th, 10:59 online

Prob. 1. Consider the problem

$$\begin{aligned} \text{minimize: } & x_1^2 + x_2^2 \\ \text{subject to: } & (x_1-1)^2 + (x_2-2)^2 \le 2 \\ & 2x_1 + x_2 \ge 3 \\ & x_1 \ge 0, x_2 \ge 0. \end{aligned}$$

(a) Derive the KKT conditions. Find a solution for these conditions. Is this the optimal solution?

(b) Does the strong duality hold?

(c) Compare the solution to the one you obtained last time using geometry. Are they the same?

(d) Consider the problem

$$\begin{aligned} \text{minimize: } & x_1^2 + x_2^2 \\ \text{subject to: } & (x_1-1)^2 + (x_2-2)^2 \le 2 + \mu_1 \\ & 2x_1 + x_2 \ge 3 + \mu_2 \\ & x_1 \ge 0, x_2 \ge 0. \end{aligned}$$

The optimal value is a function of μ_1 and μ_2, i.e., $p^*(\mu_1, \mu_2)$. Compute

$$\left.\frac{\partial p^*(\mu_1,\mu_2)}{\partial \mu_1}\right|_{(0,0)}$$
$$\left.\frac{\partial p^*(\mu_1,\mu_2)}{\partial \mu_2}\right|_{(0,0)}.$$

Solution:

(a) The KKT conditions are

$$\begin{aligned} & (x_1-1)^2 + (x_2-2)^2 \le 2 \\ & 2x_1 + x_2 - 3 \ge 0 \\ & x_1 \ge 0, x_2 \ge 0 \\ & \lambda_1 \ge 0, \lambda_2 \ge 0, \lambda_3 \ge 0, \lambda_4 \ge 0 \\ & \lambda_1((x_1-1)^2 + (x_2-2)^2 - 2) = 0 \\ & \lambda_2(2x_1 + x_2 - 3) = 0 \\ & \lambda_3 x_1 = 0 \\ & \lambda_4 x_2 = 0 \\ & 2x_1 + 2\lambda_1(x_1-1) - 2\lambda_2 - \lambda_3 = 0 \\ & 2x_2 + 2\lambda_1(x_2-2) - \lambda_2 - \lambda_4 = 0. \end{aligned}$$

There are two ways to consider this problem at this point.

A shortcut: Firstly, since the question asks you for "a" solution to the KKT condition, from strong duality, we know that this solution if exists, must be the optimal one. Using geometry, we can conjecture the solution to be $x_1 = 6/5$, $x_2 = 3/5$. Substituting these two values into the last two equations, we obtain $\lambda_1 = 0$ and $\lambda_2 = 6/5$, and into the last but two to obtain $\lambda_3 = \lambda_4 = 0$. We still need to check all other conditions are satisfied by this set of solutions, which can be verified easily. Note that here the complementary slackness condition $\lambda_1((x_1-1)^2 + (x_2-2)^2 - 2) = 0$ is satisfied with both factors being zero $\lambda_1 = 0$ and $(x_1-1)^2 + (x_2-2)^2 - 2 = 0$.

Direct solving the KKT condition:
To solve for a solution with the KKT condition, we essentially need to consider a total of 16 zero or nonzero assignments for λ_1, λ_2, λ_3, and λ_4. However, here we can separate them into several categories first with λ_3 and λ_4:

i. Suppose $\lambda_3 > 0, \lambda_4 > 0$: then $x_1 = x_2 = 0$, but this is impossible, since the first constraint is not satisfied.

ii. Suppose $\lambda_3 = 0, \lambda_4 > 0$: then $x_1 \geq 0$, $x_2 = 0$, but this is impossible, again because of the first constraint.

iii. Suppose $\lambda_3 > 0, \lambda_4 = 0$: then $x_1 = 0$, $x_2 \geq 0$. However, from the last but one condition we would have $-2\lambda_1 - 2\lambda_2 - \lambda_3 = 0$, which is impossible since $\lambda_3 > 0$ and $\lambda_1, \lambda_2 \geq 0$.

iv. Suppose $\lambda_3 = 0, \lambda_4 = 0$: this is the only possible case. Now the condition reduces to

$$
\begin{aligned}
&(x_1-1)^2 + (x_2-2)^2 \leq 2 \\
&2x_1 + x_2 - 3 \geq 0 \\
&x_1 \geq 0, x_2 \geq 0 \\
&\lambda_1 \geq 0, \lambda_2 \geq 0 \\
&\lambda_1((x_1-1)^2 + (x_2-2)^2 - 2) = 0 \\
&\lambda_2(2x_1 + x_2 - 3) = 0 \\
&2x_1 + 2\lambda_1(x_1-1) - 2\lambda_2 = 0 \\
&2x_2 + 2\lambda_1(x_2-2) - \lambda_2 = 0.
\end{aligned}
$$

Now with $\lambda_3 = \lambda_4 = 0$, we consider the assignments of λ_1 and λ_2:

i. $\lambda_1 = \lambda_2 = 0$: this is impossible since it implies $x_1 = x_2 = 0$ which violates the first condition.

ii. $\lambda_1 = 0, \lambda_2 > 0$: this will give $x_1 = \lambda_2 = 6/5$ and $x_2 = 3/5$. This is a valid solution, and thus $x^* = (6/5, 3/5)$ is optimal.

iii. $\lambda_1 > 0, \lambda_2 = 0$: from the last two conditions we then get $x_1 = \frac{\lambda_1}{\lambda_1+1}$, $x_2 = \frac{2\lambda_1}{\lambda_1+1}$. Substitute these into the first complementary slackness condition, we can solve for λ_1, which gives $\lambda_1 = \sqrt{5/2} - 1$. However, the corresponding x_1, x_2 values will violate the condition $2x_1 + x_2 - 3 \geq 0$. Thus this is also an impossible case.

iv. $\lambda_1 > 0, \lambda_2 > 0$: We would obtain a pair of solution from the complementary slackness, $(x_1, x_2) = (6/5, 3/5)$ or $(x_1, x_2) = (0, 3)$. For the former we can solve $\lambda_1 = 0$

and $\lambda_2 = 6/5$ from the last two equations in the KKT condition, but this is impossible since we had assumed $\lambda_1 > 0$. For the latter, the last but one condition would be $-2\lambda_1 - 2\lambda_2 = 0$, which also is impossible since in this case assumed $\lambda_1 > 0, \lambda_2 > 0$.

Thus overall, we found the only solution to be $x_1 = 6/5$, $x_2 = 3/5$, $\lambda_1 = 0$, $\lambda_2 = 6/5$, and $\lambda_3 = \lambda_4 = 0$.

(b) Yes it holds. This is a convex problem, and there exists a feasible point that satisfies the constraints with strict inequality, and thus Slater's condition can be used to make this assertion. Alternatively, you can find the dual problem, and show that the λ^* value we found in the previous question achieves the same dual objective value as the primal optimization problem.

(c) Yes, they are the same.

(d) Recall the local sensitivity analysis relation, and we can directly write down that

$$\left.\frac{\partial p^*(\mu_1,\mu_2)}{\partial \mu_1}\right|_{(0,0)} = -\lambda_1 = 0,$$
$$\left.\frac{\partial p^*(\mu_1,\mu_2)}{\partial \mu_2}\right|_{(0,0)} = \lambda_2 = \frac{6}{5}.$$

Note the since we need to write the second condition as $-2x_1 - x_2 + 3 \leq -\mu_2$, we need to negate the sign (λ_2 instead of $-\lambda_2$) to obtain this local sensitivity coefficient. This can be verified by considering the geometric meaning of the this perturbation.

Prob. 2. Consider the following optimization problem:

$$\begin{aligned} \text{minimize:} \quad & \sum_{i=1}^{n} \alpha_i \max\left\{\ln\frac{\sigma_i^2}{x_i}, 0\right\} \\ \text{subject to:} \quad & \sum_{i=1}^{n} \beta_i x_i \leq D, \end{aligned}$$

where $x = (x_1, x_2, \ldots, x_n) \succ 0$ are the decision variables. Assume $\sigma^2 \succ 0$, $\alpha \succeq 0$, and $\beta \succ 0$.

(a) Argue that this problem is convex;

(b) Write down the KKT conditions;

(c) Provide an explicit solution for this problem (in terms of certain intermediate variable), by providing a solution to the KKT conditions;

(d) Show that equivalently, the inequality constraint can be replaced by an equality constraint for any $0 < D \leq \sum_{i=1}^{n} \beta_i \sigma_i^2$;

(e) Let $n = 3$, $(\sigma_1^2, \sigma_2^2, \sigma_3^2) = (1, 4, 25)$, $\alpha = (1, 1, 2)$, and $\beta = (2, 1, 1)$. Consider the optimal value $p^\star$ as a function of the parameter D. Find this function in a closed form.

(f) Design an algorithm to compute the optimal solution for any given $\sigma^2 \succ 0$, $\alpha \succeq 0$, and $\beta \succ 0$, with a computation complexity at most $O(n)$.

Solution:

(a) The objective function is the sum of pointwise maximum, and thus convex.

(b) If we directly write the Lagrangian including the function $\max(\cdot,\cdot)$, the resultant function is not differentiable at some points, and we cannot apply the KKT technique without complications. Instead, we first start to rewrite the problem slightly. We can rewrite the problem into the following equivalent problem:

$$\begin{aligned} \text{minimize: } & \sum_{i=1}^{n} \alpha_i \ln \frac{\sigma_i^2}{x_i} \\ \text{subject to: } & \sum_{i=1}^{n} \beta_i x_i \le D, \\ & x_i \le \sigma_i^2, \quad i = 1, 2 \ldots, n. \end{aligned}$$

To see this equivalence, denote the optimal solution in the original problem as P_1^* and the optimal solution in the new problem as P_2^*. Clearly $P_1^* \le P_2^*$, since the new problem has a more restricted feasible set (and in fact the same objective function). To see $P_1^* \ge P_2^*$, suppose otherwise, i.e., $P_1^* < P_2^*$. This implies that the optimal solution for the original problem has some $x_i^* > \sigma_i^2$ for $i \in \mathcal{A}$. Let us construct a new solution where $\hat{x}_i = \sigma_i^2$ for $i \in \mathcal{A}$ and $\hat{x}_i = x_i^*$ for $i \notin \mathcal{A}$. Clearly this new solution satisfies all the constraints in the new problem, but it has the same objective function value as P_1^*, therefore, $P_1^* \ge P_2^*$, which gives us a contradiction. Therefore, the two problems are equivalent. With this equivalent form, we can write the KKT conditions easily:

$$\begin{aligned} \sum_{i=1}^{n} \beta_i x_i - D \le 0 & \\ \lambda_i \ge 0, & \quad i = 1, 2, \ldots, n \\ x_i - \sigma_i^2 \le 0, & \quad i = 1, 2, \ldots, n \\ \lambda_i (x_i - \sigma_i^2) = 0, & \quad i = 1, 2, \ldots, n \\ \mu \Big(\sum_{i=1}^{m} \beta_i x_i - D\Big) = 0 & \\ -\frac{\alpha_i}{x_i} + \lambda_i + \beta_i \mu = 0, & \quad i = 1, 2 \ldots, n. \end{aligned}$$

(c) If $D > \sum_{i=1}^n \beta_i \sigma_i^2$, one optimal solution is clearly $x_i = \sigma_i^2$ for $i = 1, 2, \ldots, n$; the corresponding dual variables are $\mu = 0$ and $\lambda_i = \alpha_i / \sigma_i^2$.

Let us next consider the case when $D \le \sum_{i=1}^n \beta_i \sigma_i^2$. For each $i = 1, 2 \ldots, n$, either $x_i = \sigma_i^2$ or $\lambda_i = 0$ must hold. For the former, the last condition can be satisfied by choosing a proper λ_i. For the latter, we have $x_i = \frac{\alpha_i}{\beta_i \mu}$. The optimal solution thus must have the following structure: either $(x_i, \lambda_i) = (\sigma_i^2, \frac{\alpha_i}{\sigma_i^2} - \beta_i \mu)$ or $(x_i, \lambda_i) = (\frac{\alpha_i}{\beta_i \mu}, 0)$. Note for the first case $\frac{\alpha_i}{\sigma_i^2} \ge \beta_i \mu$, i.e., $\sigma_i^2 \le \frac{\alpha_i}{\beta_i \mu}$. In other words, we have $x_i = \min\left(\frac{\alpha_i}{\beta_i \mu}, \sigma_i^2\right)$. By finding a μ value such that $\sum_{i=1}^n \beta_i \min\left(\frac{\alpha_i}{\beta_i \mu}, \sigma_i^2\right) = D$, we would be able to identify the optimal point and the optimal value.

(d) Note that when $D \ge \sum_i \beta_i \sigma_i^2$, the problem is trivial (with the solution $d_i = \sigma_i^2$) and the objective is zero, and so the problem only needs computing when $D < \sum_i \beta_i \sigma_i^2$.

Denote the optimal solution in the original problem as P_1^* and the optimal solution in the new problem as P_2^*. Clearly $P_1^* \le P_2^*$, since the new problem has a more restricted feasible set (and in fact the same objective function). To see $P_1^* \geq P_2^*$, suppose otherwise, i.e., $P_1^* < P_2^*$. This implies that the optimal solution for the original problem must have $\sum_{i=1}^n \beta_i x_i < D$, and we denote such an optimal solution as $(x_1^*, x_2^*, \ldots, x_n^*)$. There must exist a x_i^* such that $x_i^* < \sigma_i^2$ (since $D < \sum_i \beta_i \sigma_i^2$). However, we can strictly improve this solution by choosing a $\delta \leq \min(\sigma_i^2 - x_i^*, D - \sum_i \beta_i \sigma_i^2)$, and set $\hat{x}_i = x_i^* + \delta$ and other $\hat{x}_j = x_j^*$ for $j \neq i$. It is easy to see that the constraints are still all satisfied, but the objective function value strictly improves. We therefore have a contradiction. We finally conclude $P_1^* = P_2^*$ in this case.

(e) Using the solution structure in (c), we can write the optimal points explicitly

$$(x_1^*, x_2^*, x_3^*) = \begin{cases} (1, 4, 25) & 31 \leq D \\ (1, 4, D-6) & 14 \leq D < 31 \\ (1, (D-2)/3, 2(D-2)/3) & 8 \leq D < 14 \\ (D/8, D/4, D/2) & 0 < D < 8 \end{cases}$$

Substituting this into the objective function gives the piece-wise logarithm function between p^* and D.

(f) The general case is somewhat complicated, and let's focus on the case $\alpha = \beta = \mathbf{1}$. The general case can be obtained in a similar manner, and is left to the students to work out. The input to the algorithm is D and the vector $(\sigma_1^2, \sigma_2^2, \ldots, \sigma_m^2)$, the latter of which is assumed to be given in a non-decreasing order. The output is $(d_1^*, d_2*, \ldots, d_m^*)$. A pseudo-algorithm can be roughly described as follows:

i. Compute a set of thresholds $T_k = \sum_{i=1}^k \sigma_i^2 + (n-k)\sigma_k^2$ for $k = 1, 2, \ldots, n$.

ii. If $D > T_n$, then $d_i^* = \sigma_i^2$ for all $i = 1, 2, \ldots, n$, and the algorithm terminates.

iii. If $0 < D \leq T_1$, then $d_i^* = D/n$, and the algorithm terminates.

iv. If $T_{k+1} \geq D > T_k$ for some $n-1 \geq k \geq 1$, then $d_i^* = \sigma_i^2$ for $i = 1, 2, \ldots, k$ and $d_i^* = (D - T_k)/(m-k)$ for $i = k+1, k+2, \ldots, n$.

A more refined version of this algorithm does not need to compute all the thresholds T_k's upfront, but instead computes them as needed to test whether D falls between two consecutive thresholds. It is clear that even the less refined version has complexity $O(n)$. This should be contrasted with algorithms for general convex programs, which usually have much higher complexity that is also related to the accuracy of the obtained solution (which converges to the optimal solution).

Prob. 3. Strong duality holds for the problem

$$\begin{aligned} \text{minimize:} &\ -4x_1^2 + x_2^2 + x_3^2 + 2(x_1 + x_2 + x_3) \\ \text{subject to:} &\ x_1^2 - 2x_2^2 + x_3^2 \leq 1, \end{aligned}$$

although the problem is not convex (see Appendix B.1). Derive the KKT conditions. Find all solutions for the KKT conditions. Which one is the optimal value (assuming primal and dual optimal values both attained)? You may need to use a program (Matlab, Mathematica, or Python package) to plot and solve a non-linear equation; approximate solution is acceptable.

Solution: Notice that there is only one constraint. The KKT conditions will allow us to represent x_i in terms of λ.

$$x_1 = -1/(\lambda - 4), \quad x_2 = 1/(2\lambda - 1), \quad x_3 = -1/(\lambda + 1).$$

If $\lambda = 0$, then we substitute $(x_1, x_2, x_3) = (1/4, -1, -1)$ into the constraint, which is indeed valid, yielding one solution to KKT. When $\lambda > 0$, we substitute it into the constraints and make it equality, from which we can solve for λ values, and then again substitute back into the objective function. Note that you need to confirm that indeed the constraint is satisfied, and only if so, it is a valid solution to KKT. The solution for this case are $(x_1, x_2, x_3) = (-0.999, 0.111, -0.1666)$ and $(x_1, x_2, x_3) = (1.009, 0.199, -0.25)$. Since the problem is not convex, KKT conditions are necessary but not sufficient. Evaluating them gives that $(x_1, x_2, x_3) = (-0.999, 0.111, -0.1666)$ is the optimal solution.

Prob. 4. Download the CVX package (with either the Python or Matlab interface), read the user manual, and then answer the following questions. The following constraints will cause an error in CVX as they will not be recognized as valid constraints in a convex optimization problem. Convert them to valid forms, and explain your answer. You may need to read part of the user guide of CVX.

(a) norm([x + 2*y, x - y])==0;

(b) norm([max(x,1), max(y,2)]) <= 3*x + y;

(c) x^3 + y^3 <= 1; x >= 0; y >= 0;

(d) x*y >= 1; x >= 0; y >= 0.

Solution:

(a) Norm function is convex but not affine. For this constraint, we can simply use the equivalent constraints $x + 2y = 0$ and $x - y = 0$.

(b) Norm function can only have affine argument to maintain its convexity. We can replace it with

$$\mathsf{norm}([u, v]) - 3x - y <= 0; \mathsf{max}(x, 1) <= u; \mathsf{max}(y, 2) <= v.$$

(c) CVX does not view x^3 as convex since it is only convex when $x \geq 0$. We can replace it with pow_pos(x,3)+pow_pos(y,3)<=1.

(d) xy is not a concave function. We can alternatively write x >=inv_pos(y) will be recognized as a convex constraint.

Prob. 5. The gap function $\phi_r(x)$ measures the gap between the largest r component and smallest r component of the vector $x \in \mathbb{R}^n$, where $1 \leq r < n$. Consider the optimization problem

$$\begin{aligned} \text{minimize: } & ||Ax - b|| \\ \text{subject to: } & \phi_r(x) \leq \alpha. \end{aligned}$$

(a) Find an equivalent LP formulation of the problem when the norm is ℓ_1.

(b) With the same ℓ_1 norm, set

$$A = \begin{bmatrix} 1 & 2 & 3 & 6 & 9 \\ 2 & 3 & 7 & 2 & 4 \\ 4 & 2 & 2 & 5 & 1 \end{bmatrix}$$

and $b = [8, -20, 9]^T$. Use CVX to solve this problem with $\alpha = 0.1, 1, 10$, and 100, respectively, for $r = 1$, $r = 2$, and $r = 3$, respectively.

(c) Consider the same problem with the difference that the norm is ℓ_3 norm. With the same matrix A and vector b above, solve the problem with CVX for various α values, and plot the optimal value p^* as a function of α, for $r = 1$, 2, and 3.

Solution:

(a) This is equivalent to

$$\begin{aligned} \text{minimize: } & \mathbf{1}^T t \\ \text{subject to: } & -t \preceq Ax - b \preceq t \\ & \sum_{i \in \mathcal{A}} x_i - \sum_{j \in \mathcal{B}} x_j \leq \alpha, \ \forall \mathcal{A}, \mathcal{B} \subseteq \{1, 2, \ldots, n\} \text{ such that } |\mathcal{A}| = |\mathcal{B}| = r. \end{aligned}$$

(b) You can either write this problem in the equivalent form, or directly utilize CVX functions, which will recognize the convexity and structure, and perform some of the equivalent form conversion for you. For example, you can use

```
n=5;

A=[1,2,3,6,9; 2,3,7,2,4; 4,2,2,5,1];
b=[8;-20;9];

cvx_begin
    variables x(n)
    minimize norm(A*x - b,1) % objective function
    subject to                   % constraints
        sum_largest(x,1) + sum_largest(-x,1) <= 0.1
cvx_end
```

Or you can use the form

```
n=5;

A=[1,2,3,6,9; 2,3,7,2,4; 4,2,2,5,1];
b=[8;-20;9];

cvx_begin
    variables x(n)
    minimize norm(A*x - b,1) % objective function
    subject to               % constraints
        for i=1:5
            for j=1:5
                x(i)-x(j)<=0.1
            end
        end
cvx_end
```

The optimal values we thus obtained is

	$\alpha = 0.1$	$\alpha = 1$	$\alpha = 10$	$\alpha = 100$
$r = 1$	29.8905	24.3619	0	0
$r = 2$	30.081	26.0952	0	0
$r = 3$	30.081	26.0952	0	0.

Remark: the last two rows are the same, because the third largest element and the third smallest element are the same, and they cancel out each other in the objective function. CVX does not give exact zero for the last two columns due to computation precision, but they are in fact zero. To verify this, you may consider use `cvx_precision('high')` or `cvx_precision('best')`.

(c) With CVX, you only need to change the norm function to `norm(A*x - b,3)`.

	$\alpha = 0.1$	$\alpha = 1$	$\alpha = 10$	$\alpha = 100$
$r = 1$	19.5154	16.2028	0	0
$r = 2$	19.5812	16.8251	0	0
$r = 3$	19.5812	16.8251	0	0.

Prob. 6. Consider the following problem

$$\begin{aligned} \text{minimize:} \ & \sum_{i=1}^{k} \phi(|r|_{[i]}) \\ \text{subject to:} \ & r = Ax - b, \end{aligned}$$

where $|r|_{[1]} \geq |r|_{[2]} \geq \ldots \geq |r|_{[m]}$ are the numbers $|r|_1, |r|_2, \ldots, |r|_m$ sorted in decreasing order. Here matrix A is $m \times n$.

(a) When $\phi(u) = u$, reformulate the problem into an LP. The problem with this penalty function minimizes the sum of the largest k residuals. (Hint: you need the formulation in exercise 5.19.)

(b) When $\phi(u)$ is the deadzone-linear penalty function

$$\phi(u) = \begin{cases} 0 & |u| \leq a; \\ |u| - a & |u| > a, \end{cases}$$

where $a > 0$, reformulate the problem into an LP. This penalty function minimizes the sum of the largest k residuals, but with a deadzone. Recall we have discussed a problem similar to this in class.

Solution: Note here the residue is of dimension m, instead of dimension r.

(a) Using the representation in Textbook exercise 5.19, we only need to replace x by $|x|$ (or more precisely, replace $Ax - b$ by $|Ax - b|$ here). The problem can be written as

$$\begin{aligned} \text{minimize:} \ & kt + \mathbf{1}^T z \\ \text{subject to:} \ & -t\mathbf{1} - z \preceq Ax - b \preceq t\mathbf{1} + z \\ & z \succeq 0. \end{aligned}$$

(b) Recall the exercise we had in class, and combine the technique we used for the deadzone-linear penalty function and the previous question, we can write

$$\begin{aligned}
\text{minimize: } & kt + \mathbf{1}^T z \\
\text{subject to: } & -(t+a)\mathbf{1} - z \preceq Ax - b \preceq (t+a)\mathbf{1} + z \\
& t\mathbf{1} + z \succeq 0 \\
& z \succeq 0.
\end{aligned}$$